

# A Survey on Reinforcement Learning Applications in SLAM

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**Abstract**—The emergence of mobile robotics, particularly in the automotive industry, introduces a promising era of enriched user experiences and adept handling of complex navigation challenges. The realization of these advancements necessitates a focused technological effort and the successful execution of numerous intricate tasks, particularly in the critical domain of Simultaneous Localization and Mapping (SLAM). Various artificial intelligence (AI) methodologies, such as deep learning and reinforcement learning, present viable solutions to address the challenges in SLAM. This study specifically explores the application of reinforcement learning in the context of SLAM. By enabling the agent (the robot) to iteratively interact with and receive feedback from its environment, reinforcement learning facilitates the acquisition of navigation and mapping skills, thereby enhancing the robot's decision-making capabilities. This approach offers several advantages, including improved navigation proficiency, increased resilience, reduced dependence on sensor precision, and refinement of the decision-making process. The findings of this study, which provides an overview of reinforcement learning's utilization in SLAM, reveal significant advancements in the field. The investigation also highlights the evolution and innovative integration of these techniques.

**Index Terms**—Simultaneous localization and mapping; Reinforcement learning; Path planning; Loop closure detection; Active SLAM

## I. INTRODUCTION

Mobile robotics involves designing, constructing, operating, and utilizing robots to perform tasks in dynamic, non-fixed environments. These robots are usually designed to be mobile and autonomous, capable of operating without direct human control [1]. Autonomous driving vehicles are one of the specific applications of mobile robotics, focusing on developing vehicles that can navigate and operate on their own in real-world environments, such as roads and highways. These vehicles use a combination of sensors, cameras, radar, Light Detection and Ranging (LiDAR), as well as advanced algorithms to perceive their surroundings and make

decisions about how to navigate safely to their destination [2]. Simultaneous Localization and Mapping (SLAM) is a key technology in Mobile Robotics and Autonomous Driving (MRAD) [3]. SLAM enables a robot to navigate and create a map in an unknown environment by continuously observing map features to determine its own position and orientation [4, 5]. Localization enables the robot to determine its position within an environment, while mapping involves constructing a representation of the environment (the map) as the robot explores it [6]. The ability to accurately localize a robot using a map of its surroundings is essential for tasks ranging from spatial exploration to autonomous driving, as it provides the necessary information for predicting obstacle movements and determining optimal maneuvers [7, 8]. Accurate localization in robotics is a complex task due to the inherent noise in sensor measurements. Addressing outliers, occlusions, and sensor failures, as well as resolving scale differences between the map and robot motion, is essential for successful localization [4]. Additionally, detecting revisited locations, known as loop closures, presents a challenge due to perceptual aliasing and sensor limitations. It is crucial to maintain a consistent estimate of the robot's pose over time, especially in dynamic environments [6, 9]. Real-time performance is also a key consideration in ensuring effective robot navigation and localization [10, 11]. Mapping presents several challenges that require attention to ensure accuracy and reliability. One such challenge is aligning sensor measurements with map features, which demands careful consideration and precise alignment [3, 12]. Furthermore, extracting meaningful features from sensor data, such as point clouds and images, is essential for constructing an accurate map [13]. It is also crucial to accurately estimate vehicle motion during mapping and to identify revisited locations to close loops in the map [14]. Choosing an appropriate map representation, whether it's 2D grids, 3D point clouds, or another format tailored to specific mapping needs, is another key consideration. Finally,

# 強化學習在 SLAM 中的應用調查

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摘要：行動機器人技術的出現，特別是在汽車產業中的應用，為豐富使用者體驗和精準處理複雜導航挑戰，引進了一個充滿希望的時代。要實現這些進步，就必須集中技術力量，成功執行許多複雜的任務，尤其是在同步定位與對應 (SLAM) 這個關鍵領域。各種人工智慧 (AI) 方法，例如深度學習和強化學習，都是解決 SLAM 挑戰的可行方案。本研究特別探討了強化學習在 SLAM 中的應用。強化學習可讓機器人 (agent) 與環境反覆互動並從環境中接收回饋，有助於機器人獲得導航及繪圖技能，從而增強機器人的決策能力。這種方法具有多種優點，包括提高導航熟練度、增強應變能力、減少對感測器精確度的依賴，以及完善決策過程。本研究概述了強化學習在 SLAM 中的應用，其結果揭示了該領域的重大進展。本研究還強調了這些技術的演進與創新整合。

索引標題-同步定位與映射；強化學習；路徑規劃；閉路偵測；主動 SLAM

以及先進的演算法來感知周遭環境，並決定如何安全導航到目的地[2]。同步定位與繪圖 (Simultaneous Localization and Mapping, SLAM) 是行動機器人與自主駕駛 (Mobile Robotics and Autonomous Driving, MRAD) 的關鍵技術 [3]。SLAM 可讓機器人在未知的環境中，透過持續觀察地圖的特徵來確定自己的位置和方向，從而進行導航和建立地圖 [4,5]。定位功能可讓機器人確定其在環境中的位置，而繪圖功能則是在機器人探索環境時，建構環境 (地圖) 的表現[6]。對於從空間探索到自主駕駛的各種任務而言，使用周圍環境的地圖對機器人進行精確定位的能力至關重要，因為它提供了預測障礙物移動和決定最佳動作的必要資訊[7, 8]。由於感測器量測的固有雜訊，機器人精確定位是一項複雜的任務。處理離群、遮擋和感測器故障，以及解決地圖和機器人運動之間的比例差異，是成功定位的關鍵 [4]。此外，由於感知上的混淆與感應器的限制，偵測重訪位置 (即迴圈關閉) 也是一項挑戰。維持機器人姿勢的估計值長期一致是非常重要的，尤其是在動態環境中 [6, 9]。即時效能也是確保有效機器人導航及定位的關鍵考量 [10、11]。繪圖有幾項挑戰需要注意，以確保準確性與可靠性。其中一項挑戰就是將感測器的量測結果與地圖特徵對齊，這需要謹慎的考量與精確的對齊[3, 12]。此外，從點雲與影像等感測器資料中擷取有意義的特徵，對於建構精確的地圖是非常重要的 [13]。此外，在製圖過程中準確估計車輛運動，以及識別重訪位置以關閉地圖中的迴圈也非常重要 [14]。選擇合適的地圖表示方式，無論是 2D 網格、3D 點雲，或是針對特定繪圖需求量身打造的其他格式，都是另一項關鍵考量。最後

I. I

移動機器人技術涉及設計、構造、操作和利用機器人在動態、非固定的環境中執行任務。這些機器人通常被設計為可移動且自主，能夠在沒有人為直接控制的情況下運作 [1]。自主駕駛車輛是移動機器人的特定應用之一，著重於開發能夠在道路和高速公路等真實環境中自行導航和操作的車輛。這些車輛結合使用感測器、攝影機、雷達、光線偵測與測距 (LiDAR)

maintaining map consistency as new data is incorporated is important for preserving its accuracy and reliability over time [6]. These challenges drive research and innovation in SLAM algorithms, aiming to improve accuracy, robustness, and efficiency. Artificial Intelligence (AI) algorithms are integral to SLAM, as they enable robots to navigate and map their environment in real-time. Specifically, Reinforcement Learning (RL) offers promising opportunities for improving exploration, localization, and map building in SLAM applications [15]. The integration of RL in the context of MRAD aims to offer a systematic approach for robots or vehicles to acquire the skills needed for navigating environments and making informed decisions using sensor data [16, 17]. Therefore, this study explores the practical uses of RL within the context of SLAM. The key contributions are detailed as follows: • The concepts of SLAM were meticulously categorized into two distinct parts: passive and active. This nuanced classification enhances the analysis of SLAM approaches in studies, offering a clearer framework for evaluating and comparing different methodologies. • The sources of data utilized for input into the SLAM algorithm have been examined. • RL in SLAM is divided into four main categories: path planning, loop closure detection, environment exploration, obstacle detection, and Active SLAM. This classification enables researchers to explore the various applications of RL in SLAM and stimulates the development of new ideas for improvement. The rest of the paper is organized as follows: In Section 2, types of SLAM approaches are introduced. Section 3 describes RL and its operational principles. Section 4 focuses on reviewing and categorizing studies that have utilized RL in the context of SLAM. Finally, Section 5 concludes the paper.

## II. WHAT IS SLAM

The SLAM technology was first introduced at a conference in San Francisco in 1986. It combines map recognition and initialization to achieve simultaneous positioning and map creation [18]. SLAM is a collection of approaches utilized by robots to autonomously determine their location and map the surrounding environment as they traverse through it. The concept of SLAM can be further categorized into two main components: (1) localization, which involves estimating the robot's position in relation to the map, and (2) mapping, which involves reconstructing the environment using visual, visual-inertial, and laser sensors mounted on the robot [19]. In modern SLAM techniques, a graphical approach is commonly adopted, specifically a bipartite graph where nodes represent either the robot or landmark poses, and edges represent measurements between poses or poses and landmarks. Imagine a robot characterized by a state vector  $x \in \mathbb{R}^2$  that defines its position and orientation (pose). The primary aim of the SLAM issue is to determine the optimal state vector  $x^*$ , minimizing the measurement error  $e_i(x)$  weighted by the covariance matrix  $\Omega_i \in \mathbb{R}^{l \times l}$ , which accounts for the uncertainty

in pose measurements, with  $l$  representing the state vector's dimension, as illustrated in Equation (1) [20].

$$x^* = \arg \min_x \sum_t e_t^T(x) \Omega_t e_t(x) \quad (1)$$

There are two primary approaches to SLAM: Active SLAM and passive SLAM [21], which are discussed in the following.

### A. Passive SLAM approach

Passive SLAM systems do not involve navigating a robot to explore unfamiliar environments. Instead, they rely on predetermined routes or manual guidance and do not actively adapt to changes in the environment. Passive SLAM is particularly suitable for scenarios where precise robot motion is not critical, focusing instead on effective mapping and localization [22]. This approach facilitates more predictable and controlled movement, which can be advantageous in specific applications. However, it also implies that the robot may lack the ability to autonomously adjust to unforeseen changes in its environment without additional intervention [18]. Passive SLAM separates the estimation of robot motion from map estimation. Manual control or adherence to predetermined waypoints is characteristic of the robot's operation [4]. In passive SLAM, Particle Filters (PF) are commonly used to estimate robot poses and build maps. PF is a probabilistic method that represents the posterior distribution using a set of particles (samples), where each particle represents a potential robot pose and map hypothesis [23].

### B. Active SLAM approach

Active SLAM involves surveying the environment using sensors that are in motion, while simultaneously estimating the status of these sensors and constructing a map. Active SLAM setups use sensor readings as input and generate real-time decisions or actions to influence future measurements [24, 25]. Typically, it involves a three-part process [26]:

- 1) The recognition of all potential locations for exploration (ideally infinite),
- 2) The calculation of the efficacy or benefit derived from the actions that would transition the robot from its present coordinates to each of those locations,
- 3) The choice and implementation of the most advantageous course of action.

Active SLAM includes modules for planning waypoints and generating trajectories. It uses methods from information theory, optimal control theory, and RL to actively steer the robot towards its destination [4]. We discussed this approach in Section 5.3 and explored the RL application in Active SLAM.

## III. DATA SOURCE

SLAM in autonomous driving typically involves integrating data from multiple sensors to create a comprehensive

隨著新資料的加入，保持地圖的一致性對於長時間保持其準確性與可靠性是非常重要的 [6]。這些挑戰驅使 SLAM 演算法的研究與創新，目的在於改善精確度、穩定性與效率。人工智慧 (AI) 演算法是 SLAM 不可或缺的部分，因為這些演算法可讓機器人即時導航並描繪環境。具體而言，強化學習 (Reinforcement Learning, RL) 為改善 SLAM 應用中的探索、定位和地圖建立提供了大好的機會 [15]。在 MRAD 的背景下整合強化學習 (RL)，目的在於提供機器人或車輛一種有系統的方法，以獲得使用感測器資料導航環境與做出明智決策所需的技能 [16、17]。因此，本研究探討 RL 在 SLAM 背景下的實際應用。主要貢獻詳述如下：- SLAM 的概念被仔細分類為兩個截然不同的部分：被動與主動。這種細微的分類方式增強了研究中 SLAM 方法的分析，為評估和比較不同的方法提供了更清晰的框架。- 我們檢視了用來輸入 SLAM 演算法的資料來源。- SLAM 中的 RL 主要分為四類：路徑規劃、迴圈關閉偵測、環境探索、障礙偵測和主動 SLAM。此分類可讓研究人員探索 RL 在 SLAM 中的各種應用，並激發改進的新想法。本論文餘下的內容組織如下：第 2 節介紹 SLAM 方法的類型。第 3 節介紹 RL 及其運作原理。第 4 節將重點放在回顧與分類在 SLAM 背景下運用 RL 的研究。最後，第 5 節為本文的結論。

II.WSLAM

SLAM 技術於 1986 年在舊金山的一次會議上首次提出。它結合了地圖辨識與初始化，以達到同時定位與建立地圖的目的 [18]。SLAM 是機器人在穿過周遭環境時，自主決定其位置並繪製周遭環境地圖的一系列方法。SLAM 的概念可進一步分為兩個主要部分：(1) 定位，包括根據地圖估計機器人的位置；(2) 繪圖，包括使用安裝在機器人上的視覺、視覺慣性和雷射感應器重建環境 [19]。在現代 SLAM 技術中，通常會採用圖形方法，特別是雙方圖，其中節點代表機器人或地標的姿勢，而邊代表姿勢或姿勢與地標之間的量測。試想一個機器人，其狀態向量  $x \in R$  定義了它的位置和方向（姿勢）。SLAM 問題的主要目的是確定最佳狀態向量  $x$ ，最小化由共變異矩陣  $\Omega \in R$  加權的量測誤差  $e(x)$ 。

表示姿勢測量的不確定性， $l$  表示狀態向量的維度，如公式 (1) [20]。

$$x = \arg \min_x \sum_t e(x) \Omega e(x)$$

(1)

SLAM 有兩種主要的方法：主動 SLAM 和被動 SLAM [21]，以下將討論這兩種方法。

A.被動 SLAM 方法

被動式 SLAM 系統並不會導航機器人探索不熟悉的環境。相反，它們依賴於預定路徑或手動引導，不會主動適應環境的變化。被動式 SLAM 特別適用於精確的機器人動作並非關鍵的場景，而著重於有效的映射與定位 [22]。這種方法有助於提高運動的可預測性和可控性，在特定應用中具有優勢。然而，這也意味著機器人可能缺乏自主調整能力，無法在沒有額外干預的情況下適應環境中不可預見的變化 [18]。被動式 SLAM 將機器人的運動估算與地圖估算分開。手動控制或遵守預定的航點是機器人運作的特徵 [4]。在被動式 SLAM 中，粒子濾波器 (PF) 通常用來估計機器人的姿勢和建立地圖。PF 是一種概率方法，使用一組顆粒 (樣本) 來表示後期分佈，其中每個顆粒代表一個潛在的機器人姿勢與地圖假設 [23]。

B.主動式 SLAM 方法

主動式 SLAM 涉及使用運動中的感測器測量環境，同時估計這些感測器的狀態並建構地圖。主動式 SLAM 設定會使用感測器讀數作為輸入，並產生即時決策或動作，以影響未來的測量 [24、25]。一般而言，它包含三個部分的流程 [26]：

- 1) 識別所有潛在的探索位置（理想情況下是無限的），2) 計算
- 機器人從目前坐標轉移到這些位置的動作所產生的功效或好處，3) 選擇並實施最有利的動作。

主動式 SLAM 包括規劃航點和產生軌跡的模組。它使用資訊理論、最佳控制理論和 RL 的方法來主動引導機器人趨向目的地 [4]。我們在第 5.3 節討論了這種方法，並探討了主動式 SLAM 中的 RL 應用。

III.DS

自動駕駛中的 SLAM 通常涉及整合來自多個感測器的資料，以建立全面的

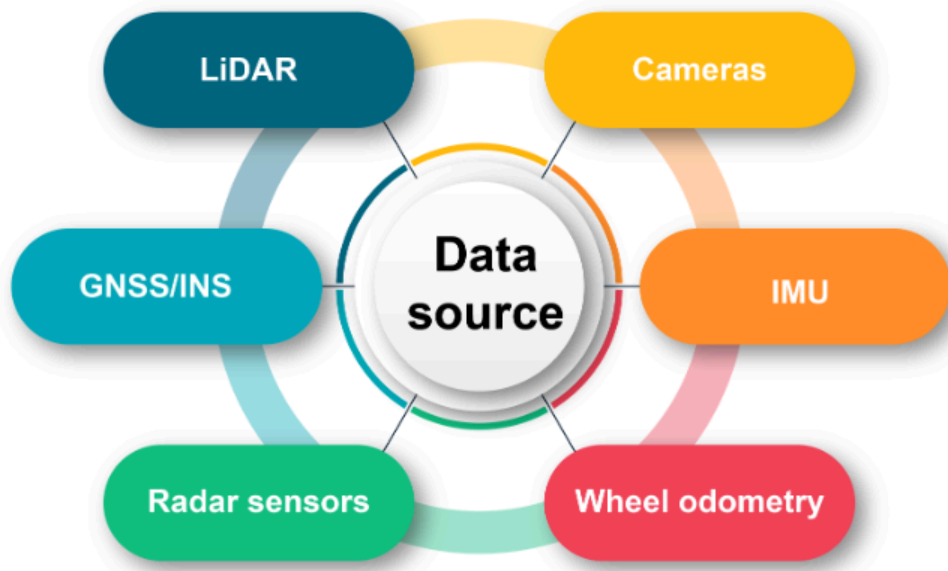


Fig. 1. Input data in SLAM.

perception of the surroundings [27, 28]. These sensors include LiDAR, cameras, GNSS/INS (Global Navigation Satellite System / Inertial Navigation System), IMUs (Inertial Measurement Units), wheel odometry, and radar. The sensor measurements are combined to provide a detailed and accurate understanding of the environment [23]. Furthermore, control commands from the vehicle's actuators such as steering, throttle, and brakes affect the robot's movement. RL techniques can optimize these control maneuvers to improve SLAM performance [4, 23]. SLAM algorithms usually start with an initial estimate of the robot's pose, derived from sources like GNSS or wheel odometry [29]. RL approaches can refine this pose estimation throughout the SLAM process, enhancing accuracy and reliability [30]. Figure 1 illustrates the input data for SLAM, offering a visual representation to aid in understanding the initial parameters of the system.

**LiDAR:** LiDAR, a remote sensing technology, uses laser light to measure distances and create detailed 3D maps of environments [31]. By emitting laser beams and calculating the time it takes for them to reflect off objects, LiDAR generates accurate point cloud data. This data is instrumental in creating comprehensive maps, identifying obstacles, and ensuring collision-free paths [32]. Additionally, LiDAR enhances precise localization by matching current scans with previously mapped features. LiDAR scans are thus pivotal for both mapping and obstacle detection, providing critical support for autonomous navigation systems [33].

**Cameras:** Cameras capture images of the surroundings, extracting visual features such as key points and edges to aid in robot localization and map construction [34]. These images are crucial for visual odometry and feature extraction, enabling the estimation of motion by tracking features across consecutive frames. Additionally, cameras help identify previously visited locations and contribute to the creation of detailed 3D maps [35].

**GNSS/INS:** GNSS offers global positioning information via satellite signals, providing data on latitude, longitude, and altitude. In contrast, INS utilizes accelerometers and gyro-

scopes to estimate position and orientation by measuring accelerations and angular rates [23]. GNSS delivers an initial position estimate, while INS ensures continuous tracking and compensates for GNSS signal outages, maintaining accurate navigation even in challenging conditions [36].

**IMU:** IMUs measure both accelerations and angular rates by combining accelerometers and gyroscopes to capture linear acceleration and angular velocity. IMUs provide short-term motion estimates, and their data is integrated with other sensors, such as LiDAR and cameras, to enhance the robustness of SLAM systems [37].

**Radar:** Radar sensors identify objects using radio waves, making them particularly valuable for obstacle detection in challenging weather conditions. In SLAM systems, radar sensors are enhancing accuracy, especially under poor lighting or occlusions [38]. The number of radar sensors used in a SLAM system varies depending on the application, requirements, and desired accuracy [39]. Some systems utilize a single radar sensor for both motion estimation and environmental mapping. This approach is particularly beneficial in low-light conditions or adverse weather conditions, where other sensors such as cameras or LiDAR may have limitations. Advanced SLAM systems may use multiple radar sensors to improve robustness against sensor failures and cover a full 360-degree field of view [40]. Combining data from multiple radars enhances accuracy and reduces blind spots [41]. Radar sensors are often integrated with other sensors to provide complementary information [42]:

- **LiDAR:** Offers long-range detection and high-resolution mapping.
- **Cameras:** Radar-camera fusion helps handle challenging lighting conditions.
- **IMU:** Improves motion estimation. Ongoing research is focused on finding optimal configurations for radar-based SLAM, balancing factors such as cost, power consumption, and sensor placement [43].

**Wheel odometry:** Wheel odometry is a fundamental localization technique that uses wheel encoders to measure the rotation of a robot's wheels, allowing it to estimate incremental movement in terms of distance and direction [44]. This method is computationally inexpensive and widely available, providing continuous pose updates as the robot moves [45]. Encoders track the distance traveled, offering incremental pose updates, but they can accumulate errors over time due to factors like wheel slippage and uneven terrain. Encoder readings are also subject to noise, which can impact accuracy [46]. In SLAM systems, wheel odometry is commonly used as one of the sensor inputs [44]. RL algorithms can learn to integrate wheel odometry information for pose estimation. In RL-based SLAM, agents are trained to fuse wheel odometry data with other sensor modalities, such as LiDAR, cameras, and IMUs, to enhance overall system robustness and accuracy [27].

#### IV. REINFORCEMENT LEARNING

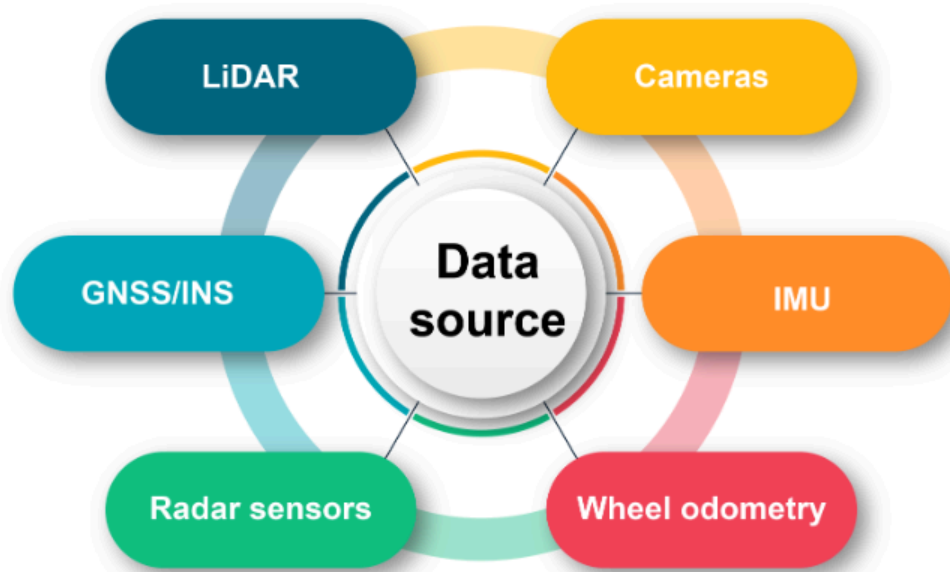


圖 1.SLAM 中的輸入資料。

感知周遭環境 [27、28]。這些感測器包括 LiDAR、攝影機、GNSS/INS ( 全球導航衛星系統 / 慣性導航系統 )、IMUs ( 慣性測量單元 )、車輪里程計和雷達。這些感測器的量測結合起來可提供對環境的詳細且精確的了解[23]。此外，來自車輛致動器的控制指令，例如轉向、油門和煞車，也會影響機器人的移動。RL 技術可以優化這些控制動作，以改善 SLAM 效能 [4、23]。SLAM 演算法通常以機器人姿態的初始估計為起點，此估計來自 GNSS 或輪子里程計等來源 [29]。RL 方法可以在整個 SLAM 過程中精細姿態估計，從而提高精確度和可靠性 [30]。圖 1 說明 SLAM 的輸入資料，以視覺化的表示方式幫助理解系統的初始參數。

**LiDAR**：LiDAR 是一種遙測技術，利用雷射光量測距離，並建立詳細的 3D 環境地圖 [31]。透過發射雷射光束並計算其從物件反射所需的時間，LiDAR 可產生精確的點雲資料。這些資料有助於建立全面的地圖、識別障礙物，並確保路徑不會發生碰撞 [32]。此外，LiDAR 還可將目前的掃描結果與先前的地圖特徵進行比對，從而增強精確定位的能力。因此，LiDAR 掃描對於地圖繪製和障礙偵測都非常重要，可為自主導航系統提供關鍵支援 [33]。

**攝影機**：攝影機捕捉周圍環境的影像，擷取視覺特徵，例如關鍵點和邊緣，以協助機器人定位和地圖建構 [34]。這些影像對於視覺測距和特徵擷取非常重要，可透過追蹤連續畫面中的特徵來估算動作。此外，攝影機有助於辨識先前造訪過的地點，並有助於建立詳細的 3D 地圖 [35]。

**GNSS/INS**：GNSS 透過衛星訊號提供全球定位資訊，提供經緯度和高度的資料。相比之下

INS 利用加速度計和陀螺儀測量加速度和角速率來估計位置和方向 [23]。GNSS 提供初始位置估算，而 INS 則可確保持續追蹤並補償 GNSS 訊號中斷的情況，即使在嚴苛的條件下也能維持精確的導航 [36]。

**IMU**：IMU 透過結合加速度計與陀螺儀來量測加速度與角速率，以擷取線性加速度與角速度。IMU 可提供短期的運動估計，其資料可與其他感測器 (例如 LiDAR 和攝影機) 整合，以增強 SLAM 系統的穩健性 [37]。

**雷達**：雷達感應器使用無線電波識別物件，因此對於在惡劣天候條件下偵測障礙物特別有價值。在 SLAM 系統中，雷達感應器可提升精確度，尤其是在光線不足或遮蔽的情況下 [38]。SLAM 系統中使用的雷達感測器數量依應用、需求及所需準確度而有所不同 [39]。有些系統利用單一雷達感應器同時進行運動估計與環境繪圖。這種方法在低光條件或惡劣天氣條件下特別有利，因為在這些條件下，其他感應器 (例如攝影機或 LiDAR) 可能會受到限制。先進的 SLAM 系統可能會使用多個雷達感應器來提高對感應器故障的穩定性，並覆蓋整個 360 度視場 [40]。結合來自多個雷達的資料可提高精確度並減少盲點 [41]。雷達感測器通常會與其他感測器整合，以提供補充資訊 [42]：

- **LiDAR**：提供長距離偵測與高解析度繪圖。
- **攝影機**：雷達與攝影機融合，有助於處理具有挑戰性的照明條件。
- **IMU**：改善運動估計。目前的研究重點在於尋找雷達式 SLAM 的最佳配置，平衡成本、功耗和感測器位置等因素 [43]。

**輪子里程計**：輪子里程計是一種基本的定位技術，它使用輪子編碼器測量機器人輪子的旋轉，讓機器人可以根據距離和方向估計增量移動 [44]。此方法的計算成本低廉且廣泛可用，可在機器人移動時提供連續姿態更新 [45]。編碼器會追蹤移動距離，提供增量姿勢更新，但由於輪子打滑和地形不平坦等因素，編碼器會隨著時間累積誤差。編碼器讀數也會受到雜訊影響，進而影響精確度 [46]。在 SLAM 系統中，車輪里程計通常用來作為感測器輸入之一 [44]。RL 演算法可以學習整合車輪里程資訊來進行姿態估計。在以 RL 為基礎的 SLAM 系統中，會訓練代理程式將輪子里程計資料與其他感測器模式 (例如 LiDAR、攝影機和 IMUs) 融合，以增強整體系統的穩定性與精確度 [27]。

#### IV.RL

To effectively analyze data and develop intelligent automated applications, a solid understanding of RL is crucial [47]. In RL, an agent learns through interactions with its environment and receiving rewards [48]. The agent explores various actions to determine which yield the highest rewards over time. Given that actions can have lasting impacts, the return value  $R$ , defined by the reward function, is computed by summing discounted future rewards across episodes, as shown in Equation (2) where  $\gamma$  denotes the discount factor, and  $r(s_t, a_t)$  represents the reward for action  $a_t$  in state  $s_t$ .

$$R = \sum_{i=0}^T \gamma^i r(s_i, a_i)$$

Each state is assigned a state value  $V(s)$ , which represents the expected return an agent can obtain by selecting actions. Equation (3) illustrates how  $V(s)$  is computed.

$$V(s) = \mathbb{E}[R|s_0 = s] \text{ or } V(s) = \mathbb{E}[r(s) + \gamma V(s_{\text{next}})] \quad (2)$$

The expectation symbol  $E$  represents the value of a state as the expected return while following a specific policy. The value of a state, denoted as  $r(s)$ , is the total expected sum of rewards achievable in that state [49]. Additionally, each action within a state has an action value  $Q(s, a)$  determined by the Bellman equation (Equation 4) [50].

$$Q(s, a) = \mathbb{E}[R|s_0 = s, a_0 = a] \\ \text{or } Q(s, a) = \mathbb{E}[r(s, a) + \gamma Q(s_{\text{next}}, a_{\text{next}})] \quad (3)$$

The state-action value, or  $Q$ -value, is the expected return of taking a specific action  $a_0$  in a state  $s_0$  and then following the policy throughout the episode. The relationship between state value and  $Q$ -value can be described by Equation (5).

$$V(s) = \sum_{a \in A} \pi(a|s) Q(s, a)$$

Given a policy  $\pi$  that selects action  $a$  in state  $s$ , a  $Q$ -table can be constructed for a Markov Decision Problem (MDP) where all states and actions are known and limited. In this process, the agent takes actions and updates the  $Q$ -value in the  $Q$ -table based on the return received from the environment. The  $Q$ -value is updated using the Equation (6), Where the updated  $Q$  value, denoted as  $Q'$ , is calculated using a learning rate represented by  $\alpha$  [48].

$$Q'(s_t, a_t) = Q(s_t, a_t) + \alpha([r_t + \gamma Q \max(s_{t+1}, a_{t+1})] - [Q(s_t, a_t)]) \quad (4)$$

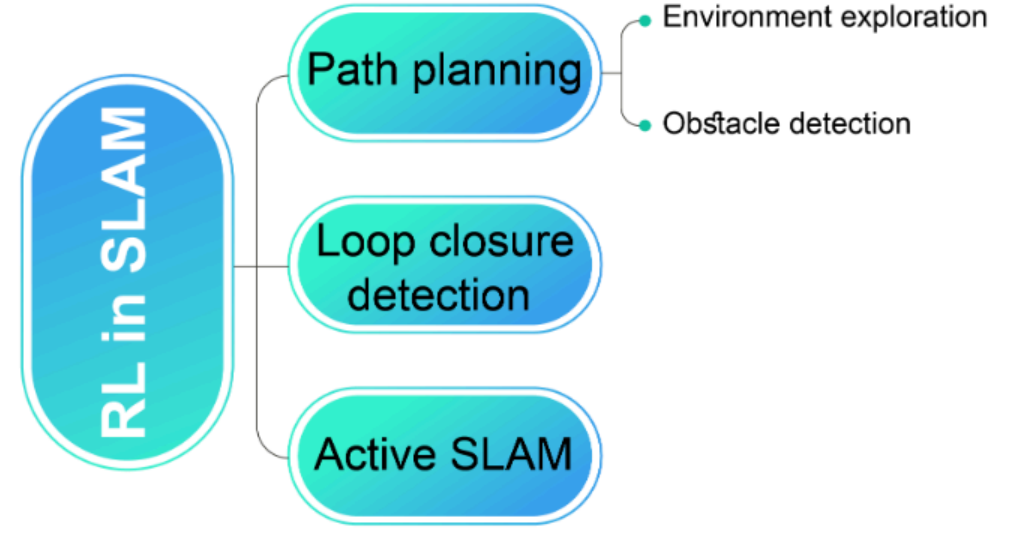


Fig. 2. Reinforcement learning.

Figure 2 illustrates an agent that learns from its environment using a RL algorithm. The algorithm updates the policy based on the actions taken, with the environment providing feedback in the form of rewards and the next state. This information is then used by the RL algorithm to improve the selection of actions.

#### A. Deep Reinforcement Learning

Deep Reinforcement Learning (DRL) is a ML approach that combines reinforcement learning with deep neural networks [51]. It differs from traditional methods by learning feature representations directly from raw data, such as images and sensor readings, rather than relying on handcrafted features. Deep neural networks, like Deep Q-Networks (DQN) or policy networks, are used to approximate the value function or policy in RL tasks [52]. This approach has shown great promise in solving complex decision-making problems and has been applied in various fields such as robotics, gaming, and autonomous systems [49, 53].

### V. CLASSIFICATION OF EXISTING METHODS

RL has many applications within SLAM, an intersection rich with research endeavors. This study reviews the diverse applications of RL within SLAM, drawing upon an array of articles for insights and analysis. These applications span various domains, including path planning, loop closure detection, and Active SLAM, as delineated in Figure 3, underscoring the multifaceted utility of RL techniques in advancing SLAM methodologies.

#### A. Path planning

Robot path planning technology, pivotal in robotics research, optimizes criteria such as minimizing work costs and finding the shortest route, ensuring efficient navigation while avoiding obstacles [54]. Path planning methods can generally be categorized into traditional and AI approaches. When considering environmental information, these methods can be further classified into global planning with known environmental data and local planning with unknown environmental

為了有效地分析資料和開發智慧型自動化應用程式，對 RL 有扎實的了解是至關重要的 [47]。在 RL 中，代理程式透過與環境互動並接受獎勵來學習 [48]。代理會探索各種行動，以確定哪些行動隨著時間的推移能產生最高的回報。鑑於行動可以產生持久的影響，回報值  $R$ （由回報函數定義）是透過將各集的未來回報折現總和而計算出來的，如公式 (2) 所示，其中  $\gamma$  表示折現因子，而  $r(s,a)$  表示行動  $a$  在狀態  $s$  的回報。

$$R = \sum_{i=0}^{\infty} \gamma^i r(s, a)$$

每個狀態都會被指派一個狀態值  $V(s)$ ，代表代理人透過選擇行動所能獲得的預期回報。  
等式 (3) 說明如何計算  $V(s)$ 。

$$V(s) = E[R|s=s] \text{ 或 } V(s) = E[r(s) + \gamma V(s)] \quad (2)$$

期望符號  $E$  代表狀態的價值，也就是遵循特定政策時的預期回報。一個狀態的值，表示為  $r(s)$ ，是在該狀態下可達到的預期報酬總和 [49]。此外，狀態中的每個動作都有一個動作值  $Q(s,a)$ ，由 Bellman 方程（等式 4）決定 [50]。

$$Q(s, a) = E[R|s=s, a=a] \\ \text{或 } Q(s, a) = E[r(s, a) + \gamma Q(s, a)] \quad (3)$$

狀態行動值（或稱  $Q$  值）是採取特定行動（state-action value）的預期回報。狀態值和  $Q$  值之間的關係可以用公式 (5) 來描述。

$$V(s) = \sum_{a \in A} \pi(a|s) Q(s, a)$$

給定一個在狀態  $s$  中選擇動作  $a$  的政策  $\pi$ ，就可以為一個所有狀態和動作都是已知且有限的馬可夫決策問題 (Markov Decision Problem, MDP) 建構一個  $Q$  表。在此過程中，代理會採取行動，並根據從環境接收到的回報更新  $Q$  表中的  $Q$  值。 $Q$  值的更新使用公式 (6)，其中更新的  $Q$  值（表示為  $Q$ ）使用以  $\alpha$  表示的學習率來計算 [48]。

$$Q(s, a) = Q(s, a) + \alpha([r + \gamma Q \max(s, a)] - [Q(s, a)]) \quad (4)$$

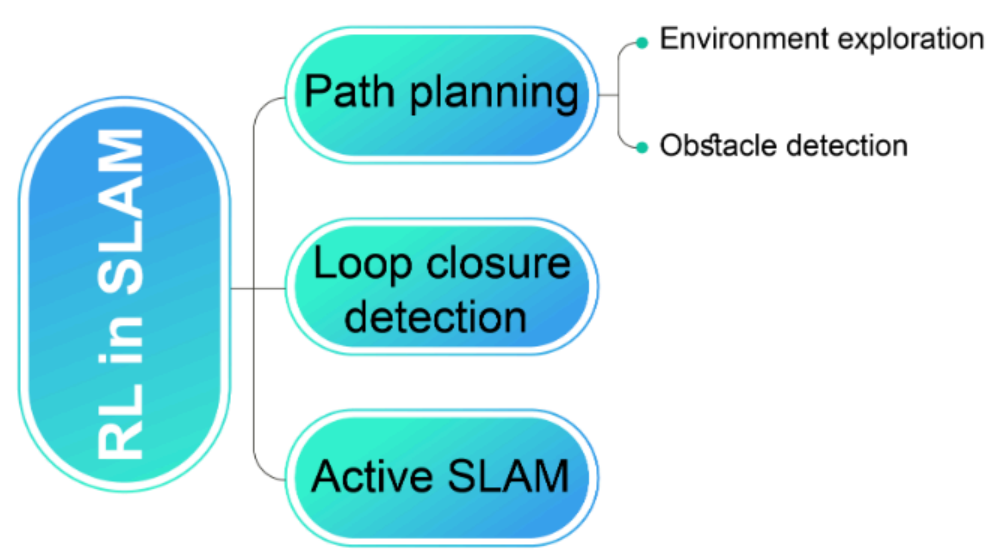


圖 2.強化學習。

圖 2 展示了一個使用 RL 演算法從環境中學習的代理程式。演算法會根據所採取的行動更新政策，而環境會以獎勵和下一個狀態的形式提供回饋。RL 演算法會使用這些資訊來改善行動的選擇。

A.深度強化學習

深度強化學習 (DRL) 是一種結合強化學習與深度神經網路的 ML 方法 [51]。它與傳統方法的不同之處在於直接從原始資料（例如影像和感測器讀數）學習特徵表現，而非依賴手工製作的特徵。深度神經網路，如深度  $Q$  網路 (Deep Q-Networks, DQN) 或政策網路，可用於近似 RL 任務中的值函數或政策 [52]。這種方法在解決複雜的決策問題上顯示了極大的前景，並已應用在機器人、遊戲和自主系統等各個領域 [49、53]。

V.現有方法的分類

RL 在 SLAM 中有許多應用，這是一個充滿研究努力的交叉領域。本研究回顧了 RL 在 SLAM 中的多種應用，並參考了多篇文章以提供深入的見解與分析。這些應用橫跨不同領域，包括路徑規劃、迴圈封閉偵測和主動式 SLAM，如圖 3 所示，突顯 RL 技術在推動 SLAM 方法上的多面性。

A.路徑規劃

機器人路徑規劃技術在機器人研究領域舉足輕重，可優化工作成本最小化和尋找最短路徑等準則，確保在避開障礙物的同時進行高效導航 [54]。路徑規劃方法一般可分為傳統方法與 AI 方法。當考慮到環境資訊時，這些方法可進一步分為已知環境資料的全局規劃與未知環境的局部規劃。

information [55]. Wang [56] developed a visual SLAM system utilizing the ORB-SLAM3 framework. The system's primary function involves the generation of a dense point cloud map. Subsequently, this dense point cloud map from the visual SLAM system is converted into an octomap, followed by a projection transformation to the grid map. The next stage involves the development of a path planning algorithm rooted in RL. Experimental comparisons were conducted among the Q-learning algorithm, the DQN algorithm, and the SARSA algorithm. The outcomes showed that the DQN algorithm exhibits the swiftest convergence rate and superior performance, particularly in intricate environments characterized by high dimensions. Nam et al. [30] introduced a novel framework for the navigation of mobile robots, integrating two established approaches (SLAM and DRL), to improve operational efficiency. The framework leverages SLAM to construct maps and pinpoint the robot's coordinates, while employing an Ant Colony Optimization (ACO) algorithm to formulate a predetermined route. In scenarios characterized by varying obstacles within the environment, the framework adopts DRL-based techniques for localized path planning. Furthermore, the suggested framework conducts a comparative analysis and assessment of the efficacy of three distinct DRL-based navigation algorithms: Deep Generative Network (DGN), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Proximal Policy Optimization (PPO).

1) *Environment exploration*: The robot exploration model integrates various exploration methods and technologies, empowering robots to autonomously navigate, map, and explore unfamiliar environments efficiently. It leverages advancements in robotics, AI, and sensor technology to seamlessly fulfill these objectives [57]. Chen et al. [55] introduced a DRL-based robot exploration model designed for navigating unknown environments without any collisions. This innovative approach integrates SLAM technology and a DRL dual-mode structure to address local-minimum issues. After 30 training rounds, the model successfully achieved zero collisions and minimized repeated exploration. It surpasses existing methods for exploring unknown environments by a margin of less than 5%. Li et al. [58] examined the concept of automatic exploration within unfamiliar environments through the application of DRL alongside a graph-based SLAM technique known as Karto SLAM. The proposed framework incorporates decision-making, planning, and mapping components that make use of a deep neural network to acquire knowledge pertaining to exploration strategies.

2) *Obstacle detection*: The obstacle detection model involves the development of algorithms and technologies to enable vehicles to detect obstacles in their surroundings accurately and in real time to ensure safe navigation and collision avoidance [59]. Wen et al. [60] used a fully convolutional residual networks method to identify road obstacles. The dueling DQN algorithm is also used in designing the robot's path. A two-dimensional map of the route is created by FastSLAM. According to Nam et al. study in [61], SLAM algorithms are effective for mapping in the environment and

DRL algorithms can find dynamic obstacles well, but SLAM algorithms and DRL algorithms alone do not perform perfectly. In this method, SLAM algorithms combine data from several sensors, including LiDAR, to make a map of the environment. The ACO algorithm is used for planning to find the shortest optimal global path. DRL algorithms help the robot in planning the local path. In this way, the robot can make decisions based on its current situation and goal. Fayjie et al. [62] used DRL for autonomous navigation and obstacle avoidance in self-driving cars. This study uses camera and laser sensor data and a trained neural network for driving. The deep Q-network approach has also been implemented for autonomous driving simulation tests.

### B. Loop closure detection

Loop closure, a crucial process in robotics and autonomous vehicles, addresses inaccuracies in sensor measurements [63]. It also tackles various issues affecting reliability, such as drifting, acceleration changes, and weather conditions. By detecting when a vehicle revisits a previously visited location, loop closure helps to correct any accumulated errors in the system's map or position estimate [64]. This process is essential for ensuring the accuracy and credibility of the vehicle's navigation system and overall performance [65]. Iqbal et al. [66] investigated loop closure detection in simulated environments using a DRL approach. In this study, training was improved with entropy maximization for batch size selection. Furthermore, Bag-of-Words (BOW) method is used for loop closure and localization in maps, which represent an image using locally created features. DRL trains the probabilistic policy for loop closure detection. Furthermore, Convolutional Neural Networks (CNN) and region-based features are used for landmark proposal and matching. In another study, Iqbal [67] presented two approaches to solve the problem of loop closure detection. The first approach uses statistical and clustering methods. In the second approach using deep reinforcement learning, loop closure detection is considered as a reward-driven optimization process. The proposed structure is implemented in a simulated grid environment. After generating the data, the learning process is done for the agent and the agent learns to detect the loop closure in variant environments.

### C. RL in Active SLAM

As mentioned in Section 2, Active SLAM is a method used by robots and autonomous systems to actively explore their surroundings. This approach allows the system to continuously enhance its understanding of the environment while also updating its position in real-time [68]. By making informed decisions on where to move next, the robot can effectively gather the most valuable data for mapping and localization purposes. This dynamic technique enables the system to adapt to changing environments and efficiently navigate through unknown areas [69, 70]. Fang et al. [71] harnessed the power of MuZero to improve agents' planning abilities for joint Active SLAM and navigation tasks. These tasks involve navigating through unfamiliar environments while creating a

當考慮到環境資訊資訊[55]。Wang [56]利用 ORB-SLAM3 框架開發了一套視覺 SLAM 系統。該系統的主要功能包括產生密集點雲圖。隨後，這個來自視覺化 SLAM 系統的密集點雲圖會被轉換成八分圖，接著再投影轉換成網格圖。下一階段則是開發一個以 RL 為基礎的路徑規劃演算法。實驗比較了 Q-learning 演算法、DQN 演算法和 SARSA 演算法。結果顯示 DQN 演算法展現出最快速的收斂速度與優異的效能，尤其是在複雜的高維數環境中。Nam 等人 [30] 針對行動機器人的導航提出了一個新穎的架構，整合了兩種既有的方法 (SLAM 與 DRL)，以提高運作效率。此架構利用 SLAM 來建構地圖並精確定位機器人的座標，同時運用螞蟻群落最佳化 (Ant Colony Optimization, ACO) 演算法來制定預定路徑。在環境中存在不同障礙物的情況下，該框架採用基於 DRL 的技術進行定位路徑規劃。此外，建議的框架還對三種不同的基於 DRL 的導航演算法的效能進行了比較分析和評估：Deep Generative Network (DGN)、Twin Delayed Deep Deterministic Policy Gradient (TD3) 及 Proximal Policy Optimization (PPO)。

1) 環境探索：機器人探索模型整合了各種探索方法與技術，讓機器人能夠自主導航、製圖並有效探索陌生環境。它利用機器人、人工智慧與感測器技術的進步，無縫地實現這些目標 [57]。Chen 等人[55] 提出了一個以 DRL 為基礎的機器人探索模型，專為在沒有任何碰撞的情況下導航未知環境而設計。此創新方法整合 SLAM 技術與 DRL 雙模式結構，以解決局部最小化問題。經過 30 輪訓練後，該模型成功實現了零碰撞和最小化重複探索。Li 等人[58] 研究了在不熟悉環境中自動探索的概念，方法是將 DRL 與稱為 Karto SLAM 的圖形 SLAM 技術一起應用。所提出的框架結合了決策、規劃與映射元件，利用深度神經網路取得探索策略的相關知識。

2) 障礙偵測：障礙偵測模式涉及演算法與技術的開發，使車輛能即時準確偵測周遭環境中的障礙物，以確保安全導航與避免碰撞[59]。Wen 等人[60] 使用全卷积残差网络方法来识别道路障碍物。決勝 DQN 演算法也用於設計機器人的路徑。由 FastSLAM 建立路徑的二維圖。根據 Nam 等人的研究[61]。

SLAM 演算法對於環境中的映射很有效，DRL 演算法也能很好地找到動態障礙物，但是單靠 SLAM 演算法和 DRL 演算法的表現並不完美。在本方法中，SLAM 演算法結合多個感測器 (包括 LiDAR) 的資料，來製作環境地圖。ACO 演算法用於規劃，以找出最短的最佳全局路徑。DRL 演算法則協助機器人規劃局部路徑。如此一來，機器人就能根據其目前的狀況與目標做出決策。Fayjie 等人 [62] 將 DRL 應用於自駕車的自主導航與避障。這項研究使用攝影機與雷射感應器資料，以及訓練好的神經網路來進行駕駛。深度 Q 網路方法也已實作於自主駕駛模擬測試。

## B.迴路閉合偵測

迴圈封閉是機器人與自主車輛的關鍵流程，可解決感測器量測不準確的問題 [63]。它還能解決各種影響可靠性的問題，例如漂移、加速度變化和天氣狀況。透過偵測車輛何時再次造訪先前造訪過的位置，迴圈封閉有助於修正系統地圖或位置估計中的任何累積誤差 [64]。這個過程對於確保車輛導航系統的精確度與可信度以及整體效能是非常重要的 [65]。Iqbal 等人 [66] 使用 DRL 方法研究了模擬環境中的閉環偵測。在這項研究中，使用熵最大化來選擇批次大小，以改善訓練。此外，Bag-of-Words (BOW) 方法用於地圖中的迴圈封閉與定位，而地圖是使用局部建立的特徵來表示影像。DRL 會訓練偵測迴圈封閉的概率政策。此外，卷積神經網路 (CNN) 和基於區域的特徵可用於地標建議和匹配。在另一項研究中，Iqbal [67] 提出了兩種解決迴圈封閉偵測問題的方法。第一種方法使用統計與聚類方法。在使用深度強化學習的第二種方法中，迴圈封閉偵測被視為獎勵驅動的最佳化過程。建議的結構在模擬網格環境中實作。在產生資料之後，會針對代理程式進行學習，代理程式會學習在不同的環境中偵測迴路閉合。

## C.主動式 SLAM 中的 RL

如第 2 節所述，主動式 SLAM 是機器人和自主系統用來主動探索周圍環境的方法。此方法可讓系統持續加強對環境的了解，同時也能即時更新其位置 [68]。透過對下一步的移動位置做出明智的決定，機器人可以有效地收集最有價值的資料，以達到繪圖和定位的目的。此動態技術可讓系統適應不斷變化的環境，並有效率地在未知區域導航 [69、70]。Fang 等人[71] 利用 MuZero 的強大功能來改善主動式 SLAM 與導航任務的規劃能力。這些任務包括在不熟悉的環境中導航，同時建立一個

map and determining the agent's location simultaneously. The paper introduces the SLAMuZero framework, which combines SLAM with the tree-search-based MuZero. SLAMuZero employs an explicit encoder-decoder architecture for mapping, along with a prediction function to assess policy and value using the generated map. The integration of SLAMuZero leads to a substantial decrease in training time. Placed et al. [68] utilized deep Q-learning architecture with laser measurements for navigation and focused on reducing uncertainty in robot localization and map representation. Trained agents reduce uncertainty, transfer knowledge to new maps and learn to navigate and explore in simulations. Pei et al. [72] introduced Active relative localization for multi-agent SLAMs. The task allocation algorithm is based on DRL and utilizes a novel Deep Q Network (MAS-DQN) to enhance collaboration efficiency in SLAM. Alcalde et al. [73] used two agents namely completeness-based and uncertainty-based agents. According to the results, these agents completed maps without collisions. The uncertainty-based agent generated longer paths but better maps, and the Active SLAM DRL solution improved performance in complex environments. Table 1 offers a comparative examination of the scrutinized research endeavors utilizing RL in the context of SLAM applications. It delves into the simulation environment, deep learning techniques, SLAM methodologies, and RL algorithms employed in these studies.

## VI. CHALLENGES IN APPLYING RL TO SLAM

High-dimensional state and action spaces, sample efficiency, and sensor/actuator delays are some of the difficulties faced by RL implementations in SLAM.

The state space encapsulates all possible states in the environment. This encompasses the intricate array of sensor data vital for SLAM operations. Meanwhile, the action space delineates the spectrum of feasible actions available to the agent [74]. Within SLAM, these actions pertain to the movements, encompassing maneuvers like turning and accelerating. SLAM systems operate amidst a milieu of high-dimensional sensor data, ranging from intricate camera images to intricate LiDAR point clouds, essential for navigating complex environments [75]. However, the efficacy of RL agents in handling such expansive input spaces is challenged by the escalating computational complexity inherent in high-dimensional realms [31].

The ability of a RL algorithm to learn from a small number of interactions (samples) with the environment is known as sample efficiency [76]. Sample efficiency is important since real-world data collecting for autonomous vehicles in SLAM can be costly and time-consuming (e.g., using laser sensors) [77].

Sensor/actuator delays epitomize the temporal gap between perceiving an event and enacting a response. This latency, inherent to the system, poses a critical challenge. In these domains, the journey from sensing to decision-making to

action execution encompasses finite intervals, demanding precise synchronization [78]. RL algorithms must grapple with these delays to orchestrate timely and precise responses, ensuring seamless navigation and operation [16]. Within the intricate landscape of SLAM, this temporal precision assumes paramount importance, since the essence of success lies in the precision of real-time processing. This crucial element not only upholds but enhances the quality of localization and mapping, but also ensures a seamless fusion of navigational prowess [31].

## VII. CONCLUSION AND FUTURE DIRECTION

SLAM is a technique used in robotics and autonomous vehicles to create a map of an unknown environment while simultaneously keeping track of an agent's location within that environment. It is a key technology for enabling MRAD to navigate and operate in real-world settings. SLAM involves the use of various sensors such as cameras, LiDAR, and odometry to gather information about the surrounding environment and then process this data to construct a map and estimate the agent's pose. In this survey, applications that have used RL in SLAM were investigated. According to the searches, the most use of RL in SLAM was in path planning, loop closure detection, environment exploration, obstacle detection, and Active SLAM. In these problems, RL helps the agent to design an intelligent map and facilitate navigation. SLAM methods can be effectively applied in MRAD, but the sensors, and environmental factors may need to be tailored to the respective application domain. In future research on the application of RL in SLAM, several key areas hold promise for advancing the state of the art:

- Adaptive sensor fusion: Combining data from various sensors, such as cameras, LiDAR, and inertial measurement units (IMUs), is crucial for achieving robust SLAM. Future work could focus on developing RL agents capable of learning how to adaptively fuse information from these different modalities. By doing so, the overall performance and reliability of SLAM systems could be significantly enhanced, particularly in diverse and dynamic environments.
- Self-Supervised learning and data augmentation: The integration of self-supervised learning (SSL) and data augmentation techniques offers substantial potential for improving RL-based SLAM, especially in MRAD applications. Leveraging large amounts of unlabeled data and generating diverse training samples can enhance the robustness and generalization capabilities of SLAM systems. Future research should explore innovative SSL strategies and data augmentation methods to maximize the efficacy of RL in SLAM.
- Knowledge transfer across environments: For RL agents to be truly effective in real-world applications, they must be able to transfer knowledge across different maps or environments. Future studies should investigate techniques such as domain adaptation and meta-learning to facilitate better generalization of RL-based SLAM systems. These approaches can enable RL agents to apply learned knowledge from one environment to another, thereby

[71]利用 **MuZero** 的強大功能來改善代理程式的規劃能力，以便同時進行主動式 **SLAM** 與導航任務的地圖與確定代理程式的位置。本文介紹 **SLAMuZero** 架構，它結合 **SLAM** 與以樹狀搜尋為基礎的 **MuZero**。**SLAMuZero** 採用明確的編碼器-解碼器 (**encoder-decoder**) 架構來進行映射，並搭配預測功能，以使用所產生的地圖來評估政策與價值。**SLAMuZero** 的整合可大幅減少訓練時間。**Placed** 等人 [68] 利用深度 **Q-learning** 架構與雷射量測進行導航，並著重於降低機器人定位與地圖表示的不確定性。經過訓練的代理可降低不確定性，將知識轉移到新地圖，並在模擬中學習導航與探索。**Pei** 等人 [72] 為多機器人 **SLAMs** 引進主動相對定位。任務分配演算法是以 **DRL** 為基礎，並利用新穎的深度 **Q** 網路 (**MAS-DQN**) 來增進 **SLAM** 的協同效率。**Alcalde** 等人[73] 使用了兩個代理體，分別是基於完整性的代理體與基於不確定性的代理體。根據研究結果，這些代理體在完成地圖時沒有發生碰撞。基於不確定性的代理程式產生了更長的路徑，但更好的地圖，主動式 **SLAM DRL** 解決方案改善了在複雜環境中的效能。表 1 提供了在 **SLAM** 應用背景下利用 **RL** 的詳細研究工作的比較檢查。它深入探討了這些研究中使用的模擬環境、深度學習技術、**SLAM** 方法和 **RL** 演算法。

## VI.應用 RL Slam 的挑戰

高維狀態與動作空間、樣本效率以及感測器/致動器延遲是 **SLAM** 中 **RL** 實作所面臨的一些困難。

狀態空間包含環境中所有可能的狀態。這包括對 **SLAM** 作業非常重要的傳感器資料的複雜陣列。同時，動作空間劃定了代理可用的可行動作範圍 [74]。在 **SLAM** 中，這些動作與移動有關，包括轉彎和加速等動作。**SLAM** 系統在高維感測器資料的環境中運作，從複雜的攝影機影像到複雜的 **LiDAR** 點雲，這些資料對於在複雜的環境中導航非常重要 [75]。然而，由於高維領域固有的計算複雜度不斷攀升，因此 **RL** 代理在處理如此龐大的輸入空間時，其效能受到挑戰 [31]。

**RL** 演算法從與環境的少量互動 (樣本) 中學習的能力稱為樣本效率 [76]。樣本效率非常重要，因為在 **SLAM** 中，自動駕駛車輛的真實世界資料收集成本高且耗時（例如，使用雷射感應器）[77]。

感測器/致動器延遲是感知事件與執行回應之間時間差距的縮影。這種系統固有的延遲帶來了嚴峻的挑戰。在這些領域中

從感測、決策到行動執行的過程包含有限的間隔，需要精確的同步 [78]。**RL** 演算法必須處理這些延遲，以協調及時且精確的回應，確保無縫導航與操作 [16]。在 **SLAM** 錯綜複雜的環境中，時間精確度具有極大的重要性，因為成功的關鍵在於即時處理的精確度。這個關鍵要素不僅能維持並提昇定位與繪圖的品質，還能確保導航能力的無縫融合 [31]。

## VII.結論與未來方向

**SLAM** 是一種用於機器人和自主車輛的技術，用於建立未知環境的地圖，同時追蹤代理在該環境中的位置。這是讓 **MRAD** 在真實環境中導航及操作的關鍵技術。**SLAM** 涉及使用各種感測器，例如攝影機、**LiDAR** 和里程計，來收集周遭環境的資訊，然後處理這些資料，以建立地圖並估計主體的姿勢。在本調查中，研究了在 **SLAM** 中使用 **RL** 的應用程式。根據搜尋結果，在 **SLAM** 中使用 **RL** 最多的是路徑規劃、迴圈關閉偵測、環境探索、障礙偵測和主動 **SLAM**。在這些問題中，**RL** 協助代理體設計智慧型地圖並促進導航。**SLAM** 方法可以有效地應用在 **MRAD** 中，但感測器和環境因素可能需要針對各自的應用領域量身打造。在 **SLAM** 中應用 **RL** 的未來研究中，有幾個關鍵領域有望推動技術的發展：- 自適應感測器融合：結合來自不同感測器（如攝影機、**LiDAR** 和慣性測量單元 (**IMU**)）的資料對於實現穩健的 **SLAM** 是至關重要的。未來的工作重點是開發能夠學習如何自適應性融合這些不同模式資訊的 **RL** 代理。如此一來，**SLAM** 系統的整體效能與可靠性將可大幅提升，尤其是在多樣化的動態環境中。- 自我監督學習和資料增強：自我監督學習 (**SSL**) 與資料增強技術的整合為改善基於 **RL** 的 **SLAM** 提供了巨大的潛力，尤其是在 **MRAD** 應用中。利用大量未標示資料並產生多樣化的訓練樣本，可以增強 **SLAM** 系統的穩健性和泛化能力。未來的研究應該探索創新的 **SSL** 策略和資料擴充方法，以最大化 **RL** 在 **SLAM** 中的效能。- 跨環境的知識轉移：為了讓 **RL** 代理在真實世界的應用中真正有效，它們必須能夠在不同的地圖或環境中轉移知識。未來的研究應該探討領域適應和金屬學習等技術，以促進基於 **RL** 的 **SLAM** 系統更好的泛化。這些方法可以讓 **RL** 代理將從一個環境中學習到的知識應用到另一個環境中，從而

TABLE I  
COMPARISON OF REVIEWED STUDIES.

|                         | Ref  | Year | Simulation Environment                                  | Deep learning method                 | SLAM method              | RL algorithm                                         |
|-------------------------|------|------|---------------------------------------------------------|--------------------------------------|--------------------------|------------------------------------------------------|
| Path planning           | [56] | 2024 | Simple maze                                             | Deep neural network                  | ORB_SLAM3                | - Q-learning<br>- DQN<br>- SARSA                     |
|                         | [30] | 2023 | - Gazebo<br>- ROS<br>- Turtle Bot                       | Deep neural network                  | SLAM-MCL                 | - Q-learning<br>- DQN<br>- SARSA                     |
| Environment Exploration | [55] | 2024 | - Gazebo<br>- ROS                                       | CNN                                  | -                        | DQN                                                  |
|                         | [58] | 2019 | ROS                                                     | CNN                                  | Karto SLAM               | DQN                                                  |
| Obstacle detection      | [61] | 2203 | - ROS2<br>DDS Communication<br>- Gazebo                 | CNN                                  | SLAM-MCL                 | - DQN<br>- PPO<br>- TD3                              |
|                         | [60] | 2020 | Gazebo                                                  | Fully convolutional residual network | FastSLAM                 | Dueling DQN                                          |
|                         | [62] | 2018 | Unity Game Engine                                       | CNN                                  | -                        | deep Q network                                       |
| Loop Closure Detection  | [66] | 2022 | Turtlebot                                               | CNN                                  | VSLAM                    | Markov Decision Process                              |
|                         | [67] | 2019 | - ZOOX<br><br>- Autonomous driving platform             | Deep neural network<br>CNN           | SLAM-MCL<br>VSLAM        | Markov decision process                              |
| Active Slam             | [71] | 2024 | Habitat                                                 | Encoder-Decoder                      | -                        | -                                                    |
|                         | [68] | 2020 | Gazebo<br>- ROS<br>- Turtle Bot                         | Deep Neural Network                  | -                        | - DQN<br>- DDQN<br>- D3QN                            |
|                         | [72] | 2020 | - ROS<br>- Telbot                                       | Deep neural network                  | ORB_SLAM                 | MAS-DQN                                              |
|                         | [73] | 2022 | - Gazebo<br><br>- ROS<br>- ROBOTIS<br>TurtleBot3-Burger | Deep Neural Network                  | Lightweight Passive SLAM | Partially Observable Markov Decision Process (POMDP) |

improving their adaptability and performance in previously unseen settings. By addressing these areas, future research can contribute to the development of more robust, efficient, and versatile RL-based SLAM systems, paving the way for advancements the navigation of MRAD.

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表一  
已審閱研究的比較。

|        | Ref       | Year | 模擬環境                                        | 深度學習方法                                         | SLAM 方法           | RL 演算法                           |
|--------|-----------|------|---------------------------------------------|------------------------------------------------|-------------------|----------------------------------|
| 路徑規劃   | [56]      | 2024 | 簡單迷宮                                        | 深度神經網路                                         | ORB_SLAM3         | - Q-learning<br>- DQN<br>- SARSA |
|        | [30]      | 2023 | - Gazebo 深度神經網路                             | SLAM-MCL - Q-learning - ROS - DQN - Turtle Bot | - SARSA           |                                  |
| 環境探索   | [55]      | 2024 | - 涼亭 CNN - DQN - ROS                        |                                                |                   |                                  |
|        | [58]      | 2019 | ROS                                         | CNN                                            | Karto SLAM        | DQN                              |
| 障礙偵測   | [61]      | 2203 | - ROS2 CNN SLAM-MCL - DQN DDS 通訊 - PPO      |                                                |                   |                                  |
|        |           |      | - 涼亭                                        |                                                |                   | - TD3                            |
|        | [60]      | 2020 | 涼亭                                          | 全卷积<br>殘留網路                                    | FastSLAM          | 決鬥 DQN                           |
| 環狀封口偵測 | [62]      | 2018 | Unity 遊戲引擎                                  | CNN                                            | -                 | 深 Q 網路                           |
|        | [66] 2022 |      | 海龜                                          | CNN                                            | VSLAM             | 馬可夫決策過程                          |
|        | [67]      | 2019 | - ZOOX                                      | 深度神經網路<br>CNN                                  | SLAM-MCL<br>VSLAM | 馬可夫決策過程                          |
| 主動式大滿貫 |           |      | - 自動駕駛平台                                    |                                                |                   |                                  |
|        | [71] 2024 |      | 棲息地                                         | 編碼器-解碼器                                        | -                 | -                                |
|        | [68]      | 2020 | Gazebo 深度神經網路                               | - - DQN - ROS - DDQN - Turtle Bot - D3QN       |                   |                                  |
|        | [72]      | 2020 | - ROS 深度神經網路                                | ORB_SLAM MAS-DQN - Telbot                      |                   |                                  |
|        | [73]      | 2022 | - 涼亭<br><br>- ROS<br>- ROBOTIS<br>烏龜機器人3-漢堡 | 深度神經網路                                         | 輕巧<br>被動 SLAM     | 部分可觀察馬可夫決策過程<br><br>(POMDP)      |

這些方法可以使 RL 代理將從一個環境中學習到的知識應用到另一個環境中，從而提高其在以前未見過的環境中的適應性和性能。透過解決這些領域的問題，未來的研究將有助於開發更堅固、更有效率、更多用途的基於 RL 的 SLAM 系統，為 MRAD 的導航發展鋪路。

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